

The Macroeconomic Impact of Digital Healthcare Infrastructure in Mexico

Martin A. Perez Rodriguez

Department of Public Policy, Liberty University

PLCY 700: Foundations of Public Policy

Dr. Scott Boykin

September 5, 2026

My research interests in public policy and economics focus specifically in helping to elevate the quality of life in emerging economies through improved, equitable access to healthcare. The analysis I present examines Mexico's ongoing healthcare reforms through two primary sources: the academic journal which I will reference is titled *The Health Economy in Mexico* by Leonardo Lomeli Vanegas and the popular press article is titled *The Rise and Fall of Seguro Popular: Mexico's Health Care Odyssey*. Both provide relevant applications of the scientific and epistemological principles presented by Karl Popper, Thomas Kuhn, and Paul Feyerabend.

Lomeli Vanegas's journal provides a strong example of the scientific process and Karl Popper's concept of falsifiable. Popper in his teachings advocated for critical rationalism, presenting forth arguments that the pursuit of truth requires rigorous empirical evidence and that scientific theories must be fundamentally falsifiable (Walliman, 2011, p. 67).

These principles are evident in Lomeli Vanegas's methodology. Asserting the need for the government to put in place a well-organized universal healthcare system, the author leverages data compiled which had been put together by the Organization for economic Cooperation and Development (OECD). As a means of demonstrating the Mexican government's historical lack of investment, he highlights measurable metrics such as health expenditures as a percentage of GDP, hospital beds per 1,000 people, and overall public health spending (Lomeli Vanegas, 2020, pp. 5-6). Because Lomeli Vanegas relies exclusively on measurable statistics, his macroeconomic hypothesis is strictly falsifiable, satisfying Popper's fundamental requirement for scientific demarcation.

Conversely, the popular press article by *Think Global Health* lacks the empirical data required by Popper. Instead, the narrative aligns with Thomas Kuhn's theory that science indeed operates within an accepted framework, or what he called "paradigm," until an anomaly or crisis forces a shift in the status quo (Walliman, 2011, p. 68). This is exactly what the article

illustrates perfectly: the historic lack of healthcare access for 53 million Mexican citizens represented a major crisis within the old paradigm (McDonald et al., 2023). The crisis triggered a paradigm shift with the creation of the program Seguro Popular. However, the subsequent 2020 administration dismantled this functioning paradigm over fraud allegations of fraud, which resulted in an anomalous crisis where the program came to a screeching halt without a substitute in place.

To dive a little deeper into this occurrence, the Mexican government had implemented public policy to introduce a new healthcare paradigm: *Seguro Popular*. It took the new process a full decade to get all 53 million previously uninsured individuals enrolled (McDonald et al., 2023). The popular press article then describes how, years later in 2020, the new administration abruptly dismantled this functioning paradigm over allegations of fraud, unjustified financial decisions, and the inadequate management of the program. This resulted in an anomalous crisis where the health program came to a screeching halt without a viable alternative or substitute solutions in place, thus throwing the paradigm back into the realm of chaos.

Finally, the popular press narrative highlights Paul Feyerabend's epistemological anarchism. Feyerabend argued that science should not be a rigid, oppressive force, but rather one of the many inputs used to understand the world. He famously asserted that "anything goes" (Walliman, 2011, p. 71). In emerging economies, leaders often lack the digital infrastructure to gather empirical facts, forcing them to rely on rhetoric and ad-hoc solutions. For example, the administration's claim that Mexico's health system will soon "level that of Denmark, Canada, and the UK" is not an empirical, falsifiable hypothesis, but a political statement designed to drive an agenda (Galeana, 2022). Furthermore, the administration's decision to rapidly import doctors from Cuba is exactly a Feyerabendian maneuver which was an improvised, non-traditional solution to an immediate crisis.

Section 2: Scientific Research Questions

To dive further into this paradigm, two specific research questions align with Lomeli Vanegas's macroeconomic foundation:

1. To what extent does implementing a centralized digital health registry correlate with a reduction in out-of-pocket healthcare expenditures for marginalized populations in emerging Latin American economies?
2. How does the standardization of clinical metric capture within a fragmented national health system impact the efficiency of public health funding allocations over a ten-year period?

These questions exemplify rigorous research methodologies as they were presented by Walliman. First, they require empirical, measurable data rather than addressing subjective, philosophical, or any type of moral evaluations regarding healthcare access. They do not ask what is inherently right or what is wrong; rather, they specifically inquire about the relationship between two trackable variables: the implementation of modern digital public infrastructure and historically available macroeconomic data.

Furthermore, according to Popper, these questions are falsifiable. An independent researcher could collect the same historical GDP data, budget allocations, and OECD metrics utilized by Lomeli Vanegas to run their own empirical analysis, potentially falsifying the hypothesis. Finally, these questions are necessary to solve the ongoing health crisis in the Mexican healthcare paradigm. The resulting empirical analysis would provide policymakers with the objective evidence required to justify the shift toward a unified, data-driven healthcare infrastructure.

Finally, these questions are necessary to solve the real ongoing health crisis in the Mexican healthcare paradigm. It has now been decades, that emerging markets have attempted to modernize by deploying bloated, fragmented Electronic Health Record (EHR) system that fail to communicate with one another. There is a huge technological leap forward as emerging markets transition away from huge non-functional digital systems into lightweight versions of

these to push these emerging markets to the standards that 1st world economies have. The findings from the research and analysis executed based on these structured questions would provide government administrators and other officials with empirical statistical data which they need to justify a shift toward a new transparent, data-driven healthcare paradigm. The implementation of this foundational data architecture would allow governments to increase institutional transparency, radically improve their healthcare metric capture, and promote the establishment of economic stability required to sustain public policies focused on health initiatives.

Section 3: Biblical Worldview and Scientific Inquiry

In the book of Genesis, chapters 1 and 2 we learn of God's systematic order when he organized and created the world. He created the light, followed by the waters and the dry ground. He then created the plants before creating all animal life (King James Bible, 1769/2012, Genesis 1). In Genesis chapter 2, it says that after he created man, He placed him in the garden to "dress it and to keep it" (King James Bible, 1769/2012, Genesis 2:15) Once all things had been created, it is written that "out of the ground the Lord God formed every beast of the field, and every fowl of the air; and brought them unto Adam to see what he would call them; and whatsoever Adam called every living creature, that was the name thereof" (King James Bible, 1769/2012, Genesis 2:19).

This theological foundation deeply influenced humanity and our early scientists. For centuries, individuals read the word and recognized that it indeed testified of a Creator who is rational and orderly with His creations on earth. It was this deep conviction that led early scientists, such as Newton and Galileo, to develop the scientific method. They operated based on the firm believe and conviction that God governs His universe through discoverable, empirical laws. Just as the first man - Adam received the instruction to observe God's animal kingdom to arrive at a proper name for each creature, these scientists and many countless

others throughout history used empirical evidence and careful observation to arrive at truth and definite conclusions.

As we look at how we apply scripture to healthcare, in the Book of Proverbs it is written that “A false balance is an abomination to the Lord, but a just weight is his delight”. God delights in truth and exactness. In ancient times, a false balance was a rigged measuring device used to cheat the vulnerable. In modern context, a fragmented health care system and non-functional digital systems process is just the same as a “false balance”. If we are to provide the best healthcare possible, we need this true balance in our hearts and in our public policies. Thus, we need to ensure we identify the right digital systems of infrastructure to ensure we have measurable, traceable healthcare metrics and KPIs for all people. Establishing these systems is an act of ethical stewardship, ensuring that emerging economies can deploy precise public policies that ultimately elevate the lives and well-being of their citizens.

References

- Galeana, E. (2022, December 26). Mexico will have a first-class public health system: AMLO. *Mexico Business News*. <https://mexicobusiness.news/health/news/mexico-will-have-first-class-public-health-system-amlo>
- Lomelí Vanegas, L. (2020). The health economy in Mexico. *CEPAL Review*, 132, 181–193. <https://www.un-ilibrary.org/content/journals/16840348/2020/132/10/read>
- McDonald, T., Touchton, M., Knaul, F. M., Arreola-Ornelas, H., & Frenk, J. (2023, November 14). The rise and fall of Seguro Popular: Mexico's health care odyssey. *Think Global Health*. <https://www.thinkglobalhealth.org/article/rise-and-fall-seguro-popular-mexicos-health-care-odyssey>
- The Church of Jesus Christ of Latter-day Saints. (2012). *The Holy Bible: King James Version*. Intellectual Reserve, Inc. (Original work published 1769)
- Walliman, N. (2011). *Your research project* (3rd ed.). SAGE Publications.